

Response to Reviewers

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Young Children's Spontaneous Comprehension of Symbol-Referent Relationships in the Graphic Domain

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We are very thankful for the second round of comments and think the manuscript has again benefited very much from this round of revisions. We would like to heartily thank the reviewers for their continuous effort and patience. Below we provide a detailed overview of how we responded with changes to the manuscript for all points that were raised (e.g. R2.4 might be read as *reviewer 2, point 4*; also see E.1. for points from the editor). Fulfillment of the general requirements are listed at the end.

Editor's Notes

Overall Evaluation

All of the reviews praise the revision and are positively inclined towards the manuscript. I am as well. One reviewer recommended that the manuscript be accepted in its present form. The other two reviewers suggest that further revision is necessary. On the basis of these thoughtful reviews, I am not presently able to accept this version of your manuscript for publication in Child Development, but I am inviting you to submit another revision, based on the reviewers' comments. To avoid a prolonged review process, I will make a more final decision on the next round. The reviews are clear, so I will not summarize them. I do want to emphasize the following points for special attention.

Response:

We are very thankful for the opportunity to improve the manuscript and the clear feedback. This is very helpful. We have continuously amended the manuscript in line with the suggestions of the reviewers.

E1.1 Shortening

Two of the reviewers suggest places where the manuscript could be shortened. In some cases, they recommend removing particular paragraphs that are more tangential to the arguments you are making. In other cases, they recommend simplifying the arguments under consideration and making them more succinct. I know that this was a concern raised in the previous round of review, but it seems like you only cut ~250 words (or roughly half a page) from the introduction. I think a little more can be cut from the introduction (and similarly, from the general discussion) for the sake of brevity. The reviewers suggest meaningful places where you can revise. There are, similarly, places where the method is repetitive and certain

parts of your analysis could go into supplemental materials. I encourage you to make such revisions to shorten the overall length of the manuscript.

Response:

Thank you! We omitted the *General Methods* section and restructured the presentation of the studies to reduce redundancy. In addition, we followed the suggestions of the reviewers and have significantly trimmed the manuscript:

- shortened introduction by about 800 words (to 1.358 from 2.156)
- shortened the methods by 272 words (to 5.148 from 5.420)
- shortened the discussion by 120 words (to 2.155 from 2.276) words

For details, please see below.

E.1.2 Improving Clarity

In doing so, the clarity of the presentation needs to be emphasized. It is important to remember that Child Development has a wide readership, and many of your readers will not be familiar with your topic. As one of the reviewers suggests, you are introducing a novel vocabulary for conceptualizing this representation, so clarity to a wide audience is important. The reviewers bring up some of these concerns in the introduction. I want to add two thoughts to this discussion:

A) Presenting the general method used in the study before each experiment makes for a difficult read. As readers have processed this material, they are then given a complex section entitled “Data handling and analysis.” The disjointed nature of the presentation makes it difficult to remember the procedure, such that when Study 1 is then presented, it is easy for me to see how a reader could get lost in understanding the study.

Response:

We followed this suggestion and the second revision does not present a general overview of the methods prior to presenting the studies. We agree that this improves flow and reduces redundancy. The methods section of study 1 now guides the reader through the entire procedure. When presenting studies 2 and 3, we only discuss the relevant differences, namely the stimuli and sample. We now properly introduce a working definition for the terms iconic and abstract and suppress other wordings such as “conceptual” when describing symbol-referent-relationships.

E.1.4 Restructuring Figures

B) The Figures are doing a lot of work. For example, Figure 2A and B, which represent your stimuli, could be separated from Figure 2C, which represents your data (this is true across all experiments). Figure 2C (and 3C, and 4c) is small and hard to read.

Response:

We followed the suggestion of the editor and have separated the illustrations of stimuli and results. The illustrations are now placed closer to where they are discussed in the manuscript

(i.e. methods and results section). Placing the plots of the developmental trajectories in a separate figure also allows for printing them larger. We agree that this makes methods and results more accessible to the reader. Thanks!

E.1.5 Restructuring Presentation of Results

The results section of Study 1 has no reference to children's performance; it simply describes a set of analyses (and again, some details can be placed in Supplemental Materials). The critically important paragraph for the results section in Study 1 is the paragraph that starts "Relative to the Representation condition..." This paragraph comes at the end of the section and only at the end of the paragraph do you inform the reader of what the results are and where they are visualized. Please always present your raw data first, and then describe what kind of model you are constructing.

Response:

We agree that the checks for fit, convergence and predictive accuracy are not exactly results in themselves but rather indicators of whether our analyses are robust and appropriate. As such they have been moved to the supplementary materials. In the revised version we briefly state that these checks indicate that interpreting the full model is appropriate and then directly present the main results. Across three studies, this also helped shorten the manuscript by a full page.

E.1.6 Presentation of Data

Finally, the reviewers raised some issues about how these data are presented and analyzed, and I encourage you to keep these comments in mind.

Response:

Thank you! Reviewer 3 (see R.3.1.) is concerned about the reliability of point-estimates for the month of success for the experimental conditions. We agree that the months-of-success we provide for each condition should not come across as the precise general age under which any comparable group of children will succeed with the same task. For example, it is conceivable that when replicating the study with a new sample, the months of success may change slightly. We now highlight that the estimated month of success for each condition should be read as a developmental window when introducing the analyses and when presenting the results. We have consistently made minor changes throughout the manuscript to make it clear that our estimates are approximations.

E.1.7 Accessibility

To conclude, this is a strong revision and good manuscript overall. However, there are definitely places where you can take a step back and clarify your presentation. Please consider revising each study with clarity and accessibility to the broad readership of the journal in mind. A reader who is unfamiliar with your results, your hypotheses, and your analysis plan needs to be able to understand the main findings. Right now, someone who is

unfamiliar with this research can reconstruct the main finding, but it takes a great deal of effort. The clarity and significance of the findings can be made more available with further revision.

Response:

We are confident that the revised manuscript is now much more accessible for a general readership and that the overall approach and findings fit the general scope of *Child Development*.

Reviewer 1

Overall evaluation

The revised paper has clarified all questions I had raised in my previous review. The revised manuscript makes a clear and compelling case for the importance of the study to advance knowledge of the symbolic capacities of young children using graphic symbols. It is my opinion that this work - a careful and extremely thorough manipulation of perceptually relevant aspects of form - lays down a 'vocabulary' of sorts, the vocabulary that children can access, and when they can access it in development, when they encounter graphic symbols or when they want to use that vocabulary to refer to the world they inhabit. I believe this re-opens a whole new field of opportunity to re-explore what it means to be symbolic in graphic (or pictorial) modalities early in life.

Response

This is very kind. Thank you so much!

Reviewer 2

Overall evaluation

I appreciate that the authors took into account all the points raised by me and the other reviewers. As I said in my previous review, despite my qualms about the theoretical framing, the dataset is beautiful and worth publishing. That said, I still have issues with the logic of the overall argument, and I will leave suggestions for improvement below. I leave it up to the authors to decide whether they want to implement these changes or not. Finally, I am looking forward to seeing the study the authors mentioned in their response, the one suggesting that children do better in the noncommunicative MTS version of this task than on the current, communicative one.

Response:

We sincerely thank the reviewer for their encouraging remarks regarding the value of our dataset and for their continued investment in helping us refine the manuscript. We appreciate all of the suggestions below and have consistently amended the manuscript accordingly. In general, thanks for this very positive review experience!

R.2.1 Shorten introduction (pages 6-8)

While the Introduction is much improved, there are still redundant passages that go beyond the point of the present study. For instance, the authors spend two pages (pp. 6–8) discussing the several ways one can reduce iconicity between cue and target. But since the authors did just that in the study, the motivation for studying these various relationships is already well-grounded in the preceding literature review (Callaghan, DeLoache, etc.). In short, I would cut the paragraphs on pp. 6–8 and go directly from “Defining the precise ages and relative developmental trajectories for children’s sensitivity to using these graphic primitives as conveying meaning and for disambiguating reference in object-choice-tasks will provide crucial insights into children’s developing symbolic competence and the interplay of communicative and analogical capacities” (p. 6) to the first paragraph on page 9, which discusses the important connection to RMTS.

Response:

Thanks for the straightforward advice! We have deleted these paragraphs accordingly and agree that it reads better.

R.2.2 Use of “abstract” and “conceptual”

The terms “abstract” and “conceptual” are used in a vague manner throughout the manuscript, sometimes disjunctively (Abstract, p. 6). Do the authors have anything to gain from using these terms? If yes, can they perhaps illustrate them, so it’s clear what they mean by “abstract”, on the one hand, and by “conceptual”, on the other? For instance, where do Gestalt-like principles sit with respect to these two dimensions?

Response:

Thank you for your careful reading of the manuscript. We agree that using both terms throughout the manuscript is not adding clarity. We now omit the use of *conceptual* entirely. Below, we would like to briefly justify the use of the term *abstract*. We now provide a short working definition of the terms *iconic* and *abstract* in the introduction. We have generally streamlined the use of terminology in the preparation of the response.

An important contribution of our manuscripts is that children were tested with various types of symbol-referent relationships - especially less iconic ones - whereas previous work largely focussed on highly iconic symbol-referent relationships. In describing our work we need to refer to a continuum spanning from highly *iconic* (i.e. direct representation, pictorial, strong overlap in form) on one side and less iconic or *abstract* on the other side. Descriptions of symbol-referent-relationships often propose a continuum from iconic (a drawing) to abstract/arbitrary/conventional symbols (letters, Arabic numerals). A strength of our work is that we present children with symbol-referent-relationships that are stripping away from being direct representations but that are still not arbitrary or conventional. They are motivated and can be used by preschoolers spontaneously.

It is not unusual to discuss iconicity as a continuum (Morris, 1946) or to discuss symbols as “exhibiting iconicity to some degree” (Winter, 2026). In the context of the manuscript, we now provide the working definition of highly *iconic* symbol-referent relationships where symbol and referent share surface-level features (i.e. have the same shape) and have a high degree of similarity, and less iconic or *abstract* symbol-referent-relationships where symbol and referent do not or not necessarily share surface-level features (i.e. have different shapes) but share structural or relational features, for example by means of analogy in size, number, orientation, etc..

The paragraph now reads:

Hence, from 36 months onward, it may be productive to investigate whether children can comprehend symbol-referent-relationships that are less dependent on iconicity. In the context of the studies presented below, we discuss various symbol-referent-relationships as exhibiting iconicity in varying degrees (Winter, Woodin, & Perlman, 2026) and place them on a continuum from iconic to abstract: when symbol and referent are very similar and specifically share surface level features such as shape, their relation is highly iconic. To the extent that symbol and referent do not share surface level features but rather structural or relational properties, they are less iconic and more abstract. While iconicity provides an entry point for symbol-grounding in graphic communication (Callaghan, 2000) much like in the gestural domain (Bohn, Call, et al., 2019; Callaghan & Corbit, 2015; Fay, Ellison, & Garrod, 2014), it is generally also possible to draw less on visual resemblance and establish reference via analogies in size, number, orientation, position or gestalt principles - all of which are serving as fundamental avenues for encoding information in the graphic domain (Cleveland & McGill, 1984; Donnelly et al., 2007; Stevenson, Alberto, Boom, & Boeck, 2014). Any symbol-referent-relationship that children can spontaneously use early may be of particular relevance for our understanding of emerging and graphical literacy and the development of graphic systems of communication.

In the interest of brevity we would like to answer the question about symbol-referent-relationships based on Gestalt-principles here: where they sit on the spectrum from iconic to abstract depends on the Gestalt-principle. The principle of closure (completing a canonical form; seeing a shape implied when only a part of it is presented) is at work in the condition *Pars Pro Toto* in study 1, which is still highly iconic as here cue and target are sharing surface features to a large extent. The conditions in study 2 (*Orientation of Feature, Relative Position*) employ different shapes for cue and target, and thereby share less surface features and could be described as being more abstract.

Morris, C. W. (1946). *Signs, language and behavior* (p. 208). New York: Prentice-Hall.

Winter, B., Woodin, G., & Perlman, M. (2026). Defining iconicity for the cognitive sciences.

R.2.3 Order of Inferences

p.4: “Retrieval then necessitates a dual inference: mapping the symbol-referent relationship - whether via deixis, iconicity, or spatio-temporal contiguity - while simultaneously recognizing the cue’s pragmatic relevance to the immediate goal. Ultimately, the task demands both a structural match between cue and target, and a pragmatic inference regarding the cue’s symbolic intent.”

Isn’t the order of the two inferences backwards? First, recognize the cue’s symbolic intent and then inferring the referent based on the available cues (deixis, iconicity, etc.)? It would also be nice to come back to this double inference in the General Discussion and discuss the young children’s failures in terms of these two inferences: Which one of them do these children fail to make in the task?

Response:

We agree that in the vast majority of cases (and in the studies we present) recognizing communicative intent precedes inference of reference and the presentation in the paragraph was not intuitive. The order of presentation has been amended accordingly. Thank you!

As the literature review highlights, young children are able to make the required pragmatic inferences with symbolic artifacts at the end of the third year. As this is established and our studies do not contribute here, it may work best to return to this in the interim discussion for study 1 summarizing the finding for *Representation*:

Children perform above chance in the Representation condition at least as early as 36 months, which is the lower limit of the age-range. Based on the literature, it is reasonable to assume that also in our setup, children would be able to solve the task at hand between the second and third birthday (Callaghan, 1999, 2000; DeLoache & Marzolf, 1992). This result establishes that the task design and setup are sufficiently clear even for the youngest children in the sample: they follow the game of hide and seek, make the pragmatic inference that their partner is intent on helping them with a graphic cue, and can match cue and target according to the most basic symbol-referent relationship, namely an iconic representation. As the pragmatics of the task are identical across conditions, delayed success with other stimuli is unlikely to be due to pragmatic inference but rather an inability to grasp the symbol-referent-relationship in question. When providing less information as in Pars Pro Toto children succeed later at 38 months. [...]

R.2.4 Rephrasing comment on naturally meaningful behaviors

p. 5: “Although infants grasp naturally meaningful behaviors like pointing and gaze by the end of their first year, their ability to exploit less transparent symbolic artifacts emerges significantly later.”

This sentence mixes up the logical order of things: It is not the case that pointing and gaze are naturally meaningful behaviors or that symbolic artifacts are less transparent in and of themselves. Instead, it is because infants interpret pointing and gaze direction as communicative signals early on that we tend to think of them as “naturally meaningful”. The same point holds for “less transparent” symbolic artifacts. Suggestion: “Although infants interpret communicative signals such as pointing and gaze direction early on, their ability to exploit symbolic artifacts emerges significantly later.”

Response:

Again, thank you for the thoughtful reading. We have amended the manuscript accordingly and the sentence now reads:

Although infants interpret communicative signals such as pointing and gaze direction by the end of their first year, their ability to exploit symbolic artifacts emerges significantly later.

R.2.5 Clarifying discussion of mapping

p. 6: “To explore the continuum from iconic to abstract symbolic relationships, researchers can systematically reduce the resemblance between cue and target. Rather than varying levels of pictorial realism (Callaghan, 2000), one can minimize similarity to isolate the conceptual foundations of the mapping.”

This sentence is hard to follow: On the one hand, varying the level of pictorial realism automatically amounts to varying similarity. On the other hand, “the conceptual foundations of the mapping” is too vague to be meaningful. I suggest clarifying this point.

Response:

We agree that this was not ideal. The paragraph has been omitted as part of the cuts from the introduction proposed above (R.2.1.)

R.2.6 Omit comment on cognitive flexibility

p. 36: “Finally, cognitive flexibility is not only key for viewing the same stimulus from different conceptual angles but also specifically for switching modes of interpretation across conditions (Cuevas et al., 2014; Willoughby, Wirth, & Blair, 2012).”

Based on the authors’ responses to my first review, “the picture that emerges is not clear and switching effects appear moderate overall”. So why do the authors talk about cognitive flexibility being key to switching modes of interpretation across conditions? It’s not as if younger children were better in Block 1 than in the subsequent Blocks, isn’t it?

Response:

We agree. The sentence was redundant and has now been deleted from the manuscript.

R.2.7 Suppress conclusory paragraph

Last but not least, I think the last paragraph of the manuscript is redundant and can be safely deleted.

Response:

Thank you for the straightforward feedback! The closing paragraph has been deleted.

Reviewer 3

Overall evaluation

I appreciate the authors' attention to issues raised in the previous round of reviews. I think the Introduction better motivates the research questions, although I continue to find that it could be more succinct and direct in motivating research questions.

Response:

We agree with the reviewer and have shortened the manuscript by deleting material from the introduction, methods and discussion. Restructuring the methods has also allowed for reducing redundancy.

R.3.1 Reframing the estimate for age of success as a developmental window

The authors have clarified how they arrived at their conclusions around the age at which children perform above chance in each condition. I appreciate the approach, however; I continue to have several concerns regarding the strength of the claims made about temporal precision and robustness, given the underlying data structure:

Given that the dataset includes only a very small number of observations per age in months (approximately two children per month). Under such sparse sampling, the estimated developmental trajectories are necessarily driven to a large extent by the model structure (e.g., assumed functional form, smoothing, and priors), rather than empirical information at each specific month of age. I know that this modeling approach pools information across ages, but I am not convinced that it supports such high-resolution inference at the level of individual months. As such, the statement that the analyses "provide very robust estimates of the age in months when group-level performance exceeds chance level" seems overstated.

Second, the use of a threshold defined by the lower bound of the 95% credible interval exceeding 50% is reasonable as a descriptive criterion, but it remains unclear whether the exact point at which the interval no longer overlaps with chance is sufficiently stable at the level of single months given the sparsity of observations. As such, I would suggest that the authors reframe the estimate as an approximate developmental window or age range, rather than a precise month. I would also ask that they clarify that robustness

Response:

We are thankful that the reviewer is generally appreciating the analytic approach. We think our analyses are ideal for our project of tracing development across a wide variety of symbol-referent relationships as it provides information of the developmental slopes and

allows more fine-grained statements about ages of success than binning children in age-groups that are used as categorical predictors. At the same time, we agree that readers should be alerted of the fact that our point estimates for any month of success should not be read as absolute, definitive thresholds, but rather as plausible anchors for developmental windows. To address this, we have made the following changes to the manuscript:

When introducing the analyses, we highlight that the months of success are estimates pointing to developmental windows:

Children as a group were considered to perform above chance only when the lower bound of the 95% CrI no longer overlapped with the 50% chance line (e.g. figures [...]). Thereby, we can estimate a month of age for each condition that anchors an approximate developmental window during which children are likely to master the symbol-referent-relationship in question. As we are testing all conditions in a within-subjects-design, these estimates are especially helpful for evaluating the relative difficulty of the different conditions.

We remind the readers of this in the first interim discussion:

As such, these findings provide estimates for developmental windows when mastery of these symbol-referent-relationships is likely to occur: around the third birthday for the highly iconic conditions and around the fourth birthday for analogies in form. Finally, the results are perfectly in line with the hypothesized developmental succession of group level success coming to pass first with Representation, then Pars Pro Toto, Simple Form Analogy and Complex Form Analogy.

and when summarizing our contribution:

Using Bayesian GLMMs, we provide estimates in months of ages anchoring developmental windows during which children as a group are likely to perform above chance with each symbol-referent relationship.

In the final discussion, we formulate the methodological caveat that larger sample sizes (and sampling from different populations) would allow for more qualified statements:

Another general problem is that the project aims to establish developmental trajectories but does so only in a single, industrialized population (Henrich, Heine, & Norenzayan, 2010) and with the relatively small number of about two participants for any month of age. Future work needs to address the role of enculturation and specifically the influence of permanent exposure to symbolic and pictorial illustrations, writing and numbers or depictions featuring canonical shapes - ideally doing so both by investigating a larger number of children drawing from various

cultural settings, as well as less industrialized communities (Callaghan et al., 2012; Dehaene, Izard, Pica, & Spelke, 2006).

Furthermore, we have consistently introduced approximative modifiers throughout the text (e.g. children succeed around 48 months) to ensure the temporal precision of our claims is attenuated.

Finally, we suppress the term “robust” and other evaluations when describing our analyses and results, for example:

In Study 1, we found that in Representation [...] children are ~~robustly~~ better than chance at the lower boundary of our age-window.

R.3.3 Rephrasing paragraph on age and sample size

In terms of sample size, the authors the following: first, that it is unusual for age to be used as a continuous measure in developmental studies; and two, that 24 children per year is consistent with conventions in the field. I think the first statement is not accurate as many studies use age a continuous measure and moreover, I don't think the statement provides much rationale for their sample size. As for the second statement, it is unclear to me which “conventions” the authors are referencing. I don't have concerns per se about the overall sample size but do think a more appropriate rationale could be provided.

Response:

The methods section has been restructured: we omit the general methods section, provide all necessary information in presenting study 1 and then refer to it when introducing studies 2 and 3. The statement about conventions in the field has been deleted. We still would like to leave in the neutral mentions of continuous sampling in the abstract:

Participants were 304 German-speaking children continuously sampled from age 3 to 7 (Study 1, N = 106, 51 female; Study 2, N = 99, 49 female; Study 3, N = 99, 55 female).

at the end of the introduction:

The present work provides a systematic investigation into the diverse conceptual dimensions along which graphic symbols can be imbued with meaning and traces development continuously through the pre-school years employing the symbolic object-choice task as one coherent and simple paradigm.

in the methods when describing the sample:

In order to continuously trace the development of symbolic competences across the preschool years, we aimed to test a minimum of two children per month of age between the third and the seventh birthday for a total of 96 participants. [...] A sample of 106 children (M = 59.18 months, SD = 13.58 months, range 36 - 83 months; 51 female) participated in Study 1. [...]

and when summarizing the studies in the discussion:

Whereas previous work on graphic communication has focused almost exclusively on iconicity as a way of creating meaning and mostly compared performance across binned age-groups, the three studies presented here map development continuously using a highly simple and streamlined procedure across various conceptual dimensions.

In the manuscript, we do not claim that it is unusual to use age as a continuous measure in developmental psychology in general, but it is less frequent for the literature on graphic communication, where there are almost no instances of sampling continuously, let alone using age as a continuous variable in analyses, and this was a major motivation for doing this project. The only exception we came across while working on this over a couple of years is Callaghan (2000) where an ANOVA as main analysis with binned age-groups, is corroborated by a linear regression using age in months continuously.

Callaghan, T. C. (2000). Factors affecting children's graphic symbol use in the third year: Language, similarity, and iconicity. *Cognitive Development*, 15(2), 185-214.

R.3.4 Suppress “shift”

Finally, I continue to be concerned about the term “shift”. I appreciate that there is convergence around age 4 in success but this does not necessarily mark a shift, which is a theoretically laden term suggesting that abilities appear abruptly.

Response:

We agree with the reviewer that the term "shift" can imply an abrupt, discontinuous developmental leap, which does not accurately reflect the gradual, continuous nature of symbolic development in the preschool years. To align our phrasing with this point, we have removed the word "shift" in the abstract and throughout the manuscript. At the same time it is very interesting that we do not find a stepwise progression (e.g. first orientation, then size, then number) but rather that across various conditions, children appear to master them in the same age-window. To express this, we now use the more neutral phrase “convergent development” in line with the wording of the reviewer.

For example in the abstract:

Across conditions involving abstract or conceptual correspondences, results indicate converging development in symbolic competence around age 4.

and the discussion:

For conditions that relied strongly on children to abstract from surface level features and focus on analogies in form, orientation, size and number, there is no strong evidence for sequences in the ages of success. Rather, children appear to master all

symbol-referent relationships in the same age-window, which is indicative of convergent development in pragmatic and analogical reasoning skills in the fourth year.

General Requirements

A Lay summary is provided under the headline “lay summary” before the introduction:

Young children are skilled at understanding spoken language and gestures, but much less is known about how they make sense of graphic communication. We explored how children aged 3 to 7 interpret unfamiliar graphic displays in a picture-book-style game of hide and seek where they had to choose the correct hiding place based on a drawn cue. Across three studies, we tested several symbol–object relationships, ranging from clear representations to less iconic cues based on shape, size, number, orientation or spatial arrangement. Children followed representational drawings at three, but the understanding of more abstract symbols emerged around the fourth birthday. These findings show how children gradually learn to find meaning in increasingly complex visual symbols prior to literacy.

Short alternative-text description

Texts are provided with each figure and marked as “Alt-Text Figure X”

SRCD sociocultural policies

The manuscript is compliant with the SRCD sociocultural policies. Under the *General Methods* section (p.10) we report (1) the dates of data collection; (2) race/ethnicity, socioeconomic status, language, (3) place(s) from which that sample was drawn, including country, region, and report the (4) selection and recruitment procedures.

StatCheck

Both manuscript and supplement have been run through StatCheck. The main manuscript does not contain significance tests. All exploratory t-tests reported in the supplements have been evaluated as “consistent” by StatCheck.

Exploratory vs. Confirmatory Declaration

In the Methods for study 1 under Data Handling and Analyses we outline that all analyses and hypotheses were preregistered. We highlight where our analyses deviate from the preregistered analysis (i.e. comparing models via ELPD differences). Exploratory analyses are clearly discussed as such and are presented only in the supplements. We highlight that we had hypotheses about the specific order of the age of success for the experimental conditions but did not make point predictions about the month of success.

Conflict of Interest Forms.

The authors declare no conflict of interest. The respective forms are submitted along with the revision.

Open Science Declaration.

All data, analysis scripts and materials are publicly available. This open science declaration will be included on the title page:

Scientific Integrity and Openness: The data and code necessary to reproduce the analyses presented here are publicly accessible, as are the materials necessary to attempt to replicate the findings. Analyses were pre-registered. For data, analyses materials, and preregistrations see: [LINK MASKED FOR REVIEW]